

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

BIO-INFORMATICS [CSE_4012]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the concept of bioinformatics to demonstrate its scope. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Analyse the 5 steps involved in the BLAST approach. (5)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Apply the concept of tandem repeats (like VNTRs) and state how the dot plot will look when the sequence is plotted against itself. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Apply the concept of iterative alignment and along with a diagram state the process involved in PRN, the double nested iteration process. (5)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Apply the concept of profiles & HMMs in analyzing the steps in the construction of Position-Specific Scoring Matrices (PSSMs). (5)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Consider two DNA sequences have been aligned, and the following proportions of nucleotide substitutions are observed: proportion of transitions = 0.12, proportion of transversions = 0.08. Using the Kimura Two-Parameter model, evaluate the evolutionary distance d_{AB} between the sequences. (5)
Course Outcome: "CO4" / Cognitive Level: "Evaluate"
- 4) Assess the computational complexity of Quartet puzzling heuristic methodology by showing a schematic illustration of quartet puzzling in deriving a consensus tree by combining multiple quartet trees. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Evaluate"
B) Analyze the steps for implementing Monte Carlo algorithm to escape local minima. (5)
Course Outcome: "CO5" / Cognitive Level: "Analysis"
- 5) Analyze the steps involved in homology modelling for predicting the structure of a target protein. (5)
A) Course Outcome: "CO5" / Cognitive Level: "Analysis"
B) (5)

To measure the shared information content between two biological sequences, S_a and S_b , entropies are calculated based on their symbol frequencies:

- Information content of S_a : 2.5 bits
- Information content of S_b : 2.8 bits
- Conditional information content:
 - $I(S_a | S_b) = 1.2$ bits
 - $I(S_b | S_a) = 1.1$ bits

Evaluate the mutual information between S_a and S_b , and interpret its significance in the context of sequence similarity.

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	3	15(30%)
5	Create	3	15(30%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)